

Exp-14

Pre-trained CNN model as feature Extractor using Transfer Learning models

Aim :-

To implement a pre-trained CNN model as feature Extractor using transfer learning for efficient image classification

Objective :-

- ① To load and understand a pre-trained CNN architecture
- ② To freeze convolutional layers and use them as fixed feature extractors
- ③ To attach new fully connected layers for classification of new dataset
- ④ To train only the newly added layers while preserving the pre-trained features
- ⑤ To evaluate the model and visualise performance using epoch vs loss plot

Epoch

Loss

10

1.1048

20

1.0101

30

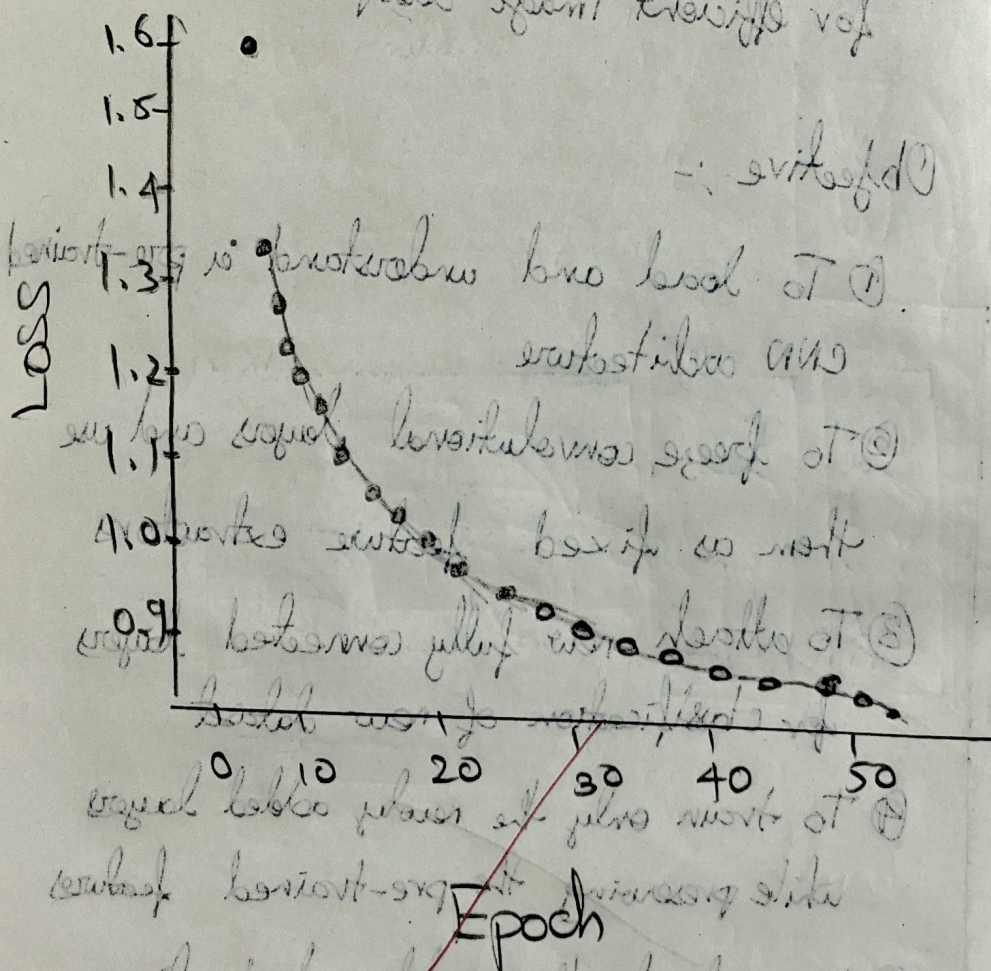
0.9412

40

0.8885

50

Q. 8487



Pseudocode:-

- ① Import necessary libraries
- ② Load dataset and preprocess it
(normalize and one-hot encode)
- ③ Load pre-trained CNN model
without top layers
- ④ Freeze all convolutional layers
to prevent retraining
- ⑤ Add new custom layers for classification
- ⑥ Compile the combined model with optimizer
and loss function
- ⑦ Train the model and record loss values
- ⑧ Print loss for epochs
- ⑨ ~~plot Epoch vs loss graph~~

END

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Result

The code executed successfully